

Junaid Ahamed

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SUMMARY

Computer Science Engineering student with experience in **backend development**, **machine learning**, and **embedded systems**. Skilled in designing **scalable APIs**, **database-driven applications**, **AI-powered solutions**, and **embedded monitoring systems** using **JavaScript**, **Python**, **PostgreSQL**, **Redis**, **React**, and **ESP32**. Strong foundation in **algorithms**, **databases**, **software architecture**, and **system design**, with practical experience delivering **production-oriented software** through internships, technical projects, and collaborative development environments.

EDUCATION

Vellore Institute of Technology, Vellore Campus	2024 - Present
- BTech in Computer Science Engineering. CGPA: 8.81 out of 10.00	Semester 5
Saint Francis DeSales School, Kerala	2022 - 2024
CBSE Class XII: A1 in All Subjects JEE Mains 2024: 97.08 Percentile National Defense Academy Exam 2024: Rank 3289 (Out of 800,000)	

EXPERIENCE

Mozilla Firefox Club, VIT Vellore	VIT Vellore
Software Development and Technical Management Team	Sep 2024 – Nov 2025
- Designed and executed technical workshops, hackathons, and developer-focused events, promoting knowledge sharing and increasing participant engagement within the university tech community through hands-on learning experiences. Coordinated with multidisciplinary teams to manage event planning, technical infrastructure, and operational workflows, ensuring smooth execution of high impact technical initiatives while maintaining effective resource coordination. - Formulated software solutions and automation tools to support event operations, streamline processes, and improve participant experience across multiple club activities enhancing scalability and efficiency for overall event management.	
Docwo Pvt. Ltd.	Ernakulam, Kerala
Back-End Development Intern	May – July 2026
- Architected and deployed core backend services for an ambulance dispatch platform (Node.js, Express, PostgreSQL, Prisma, Redis), supporting patient, driver, and admin applications with real-time ambulance discovery, booking, driver assignment, and live GPS. - Hardened the platform with JWT authentication (refresh token rotation), Google OAuth 2.0 integration, role-based access control, rate limiting on critical endpoints, and AES 256 GCM encryption for PII fields and progressive OTP lockout with attacker detection. - Built a comprehensive admin dashboard with real time KPIs, growth reports, support tickets, and full CRUD; engineered automatic driver matching (proximity search, accept/reject with timeout retry, escalation), Socket.IO live tracking, transactional notifications, Swagger API documentation, and production deployment on Azure App Service.	

PROJECTS

Emotion-Based Song Recommendation System	Python, Machine Learning, MediaPipe, TensorFlow
- Developed a real time emotion detection system using facial landmark mapping via webcam input. Trained a Machine Learning model to classify emotions based on facial feature orientation. - Integrated user input (artist, genre) to dynamically generate emotion-aware YouTube search queries.	
Real-Time Keystroke Analysis and Typing Performance System	React.js, JavaScript, CSS, HTML5
- Built a real-time typing test engine with WPM and accuracy calculation using character-level comparison logic. - Engineered state-driven timer system with controlled input handling and event-based test lifecycle management. - Created a modular paragraph system supporting multiple difficulty levels and structured content objects. - Optimized user interaction with keyboard shortcuts, focus management, and responsive UI components.	
Environmental Monitoring and Hazard Detection System	ESP32, Embedded C, Sensors
- Engineered an ESP32-based environmental monitoring system for real-time tracking of air quality, temperature, and humidity. - Implemented automatic MQ-2 sensor calibration and configurable safety thresholds, enabling dynamic user-defined limits. - Implemented a multi-level hazard detection and alert system using LEDs, buzzer, and LCD displays for real-time safety warnings.	

ACHIEVEMENTS

3 rd Place – Riddler CTF (Computer Society of India), among 750+ participants	Apr 2025
3 rd Place – Where's The Flag CTF (IEEE Computer Society), among 35+ teams	Dec 2024

SKILLS

- Programming Languages**: Python, C, C++, Java, Assembly, MATLAB
- Frontend Development**: HTML5, CSS, JavaScript, React.js
- Databases**: MySQL, PostgreSQL
- Core CS**: Data Structures & Algorithms, Design & Analysis of Algorithms, Database Management, Digital Systems Design
- Tools**: Bash, Git, GitHub, Embedded C, Prisma